

Introduction to Functional Equations: A Handful of Standard Tricks

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1 General Tips

Before we dive into the maths, here are general tips when approaching FEs:

- Before you begin attacking the FE, take a moment to **find all polynomial solutions**. (First, determine the degree, then determine the coefficients.) Knowing which solutions to expect can help motivate your solving process.
- **Pay attention to the domain/codomain of f** . Do not make the easily-avoidable (but very costly!) mistake of plugging in 0 into f when the domain of f is strictly positive. Furthermore, you may want to employ different techniques for FEs in $\mathbb{Z}$ or $\mathbb{Q}$ – more on this to come in the future.
- In your write-up for an FE problem, don't forget to **verify that your solutions work** – else, you risk losing a point!

2 Standard Tricks

The key to solving FEs is **SUBSTITUTION**.

The following points may motivate your substitutions:

- Plug in special values that result in a nice equation
e.g. $x = 0$ or $x = f(y)$
- Find special values of the function e.g. $f(0)$ or $f(1)$
- Forced cancellation
- Exploit injectivity/surjectivity
(can be established especially when there are “free” variables)
- Substitute $(x, y) = (y, x)$ to exploit “bumps” in symmetry
- Prove the function is odd/even
- Analyse pre-images e.g. a such that $f(a) = 0$
(especially when you have surjectivity)
- Change the function you are looking at
e.g. to prove $f(x) = cx$, try proving $g(x) = f(x)/x$ is constant
- fff trick: If you know what $f(f(x))$ is, apply it to $f(f(f(x)))$ in two different ways

3 The Pointwise Trap

Consider the following problem:

Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x \in \mathbb{R}, f(x)^2 = xf(x)$.

The given equation is equivalent to $f(x)(f(x) - x) = 0$.

Does this mean the solution is either $f(x) = 0 \ \forall x$ or $f(x) = x \ \forall x$?

Spoiler: No!

4 Problems

1. Find all functions $f : \mathbb{R}_{\neq 0} \rightarrow \mathbb{R}$ such that $\forall x \in \mathbb{R}_{\neq 0}$,

$$f(x) - xf\left(1 - \frac{1}{x}\right) = 1.$$

2. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(f(x)) = x^2 - x + 1$. Evaluate $f(0)$.

3. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$,

$$f(f(x) + y) = f(x^2 - y) + 4f(x)y.$$

4. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$,

$$f(f(y + f(x))) = f(x + y) + f(x) + y.$$

5. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$,

$$f(x^2 - y^2) = (x - y)(f(x) + f(y)).$$

6. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$,

$$f(f(x) + y) = f\left(\frac{x^2 + f(x)}{2}\right) + yf(x).$$

7. (MOP 2013) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$,

$$f(f(x + y)f(x - y)) = x^2 - yf(y).$$

8. (Singapore 1999) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$,

$$(x - y)f(x + y) - (x + y)f(x - y) = 4xy(x^2 - y^2).$$

9. (Japan 2024) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$,

$$f(f(x) + xy) = f(x)f(x + y).$$

10. (IMOSL 2002 A1) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$,

$$f(f(x) + y) = 2x + f(f(y) - x).$$